SteadySense

An Intelligent Real-Time Elder Care Application

Semester Project Report | **Submitted by:** Sachal

Google Drive:
`https://drive.google.com/drive/folders/1o-FFuXXVlGR8WMzkKdZqUYQyabUkKMyh?usp=sharing`

APKPure: `https://apkpure.com/p/com.example.steadysense`

Privacy Policy: `https://delicate-froyo-10153b.netlify.app/`

Research Article: `resna.org/sites/default/files/conference/2017/sdc2016/Patel.html`

Abstract

With the rapid increase in the aging population, ensuring the safety and well-being of elderly individuals has become a paramount concern. **SteadySense** is an intelligent, real-time elder care mobile application designed to bridge the gap between seniors and their families. By leveraging advanced smartphone sensors, live location tracking, and artificial intelligence, SteadySense acts as a 24/7 digital companion. It provides peace of mind to caretakers and independence to elders without requiring expensive external hardware.

1 Introduction

The objective of this semester project is to develop a comprehensive mobile solution for elder care. Traditional alert systems require users to manually press a button during an emergency, which is often impossible after a severe fall. SteadySense addresses this by continuously monitoring background sensor data to automatically detect falls and alert designated caretakers. Furthermore, it incorporates an AI voice assistant to aid elders in daily tasks and provides a real-time dashboard for caretakers to monitor their loved one's health status.

2 Key Features

SteadySense provides a robust suite of life-saving and monitoring features:

- **Automatic Fall Detection:** The application utilizes the device's built-in accelerometer running efficiently via Android Foreground Services. If a sudden impact followed by inactivity is detected, it automatically triggers a high-priority emergency sequence.
- **Live Emergency Location Tracking:** During an emergency, caretakers instantly receive push notifications with the elder's exact GPS location. The app utilizes MapLibre GL for fast, interactive map rendering and a dual-provider fallback system to ensure high accuracy.
- **Real-Time Health & Activity Sync:** Caretakers are provided with a live dashboard displaying their elder's daily step count, simulated pulse, and current activity status (e.g., Resting, Walking).
- **AI Voice Assistant:** Integrated with the Google Gemini API, the app provides a conversational companion capable of speaking both English and Urdu, catering to a diverse demographic.
- **Secure Caretaker Linking:** A robust 6-digit identification system ensures that health data and emergency alerts are only shared with explicitly authorized family members or nurses.

3 Technologies Used

The development of SteadySense involved modern software engineering practices and the following technology stack:

- **Platform:** Android SDK (Java)
- **Architecture:** Service-driven architecture utilizing Foreground Services, WakeLocks, and Broadcast Receivers to bypass Android battery optimizations during critical monitoring.
- **Backend & Database:** Firebase Authentication and Cloud Firestore. The app leverages Firestore Snapshot Listeners for real-time, low-latency UI updates.
- **Mapping Interface:** MapLibre GL for rendering customizable, high-performance local maps without relying heavily on Google Play Services intents.
- **Artificial Intelligence:** Google Gemini API for natural language processing and voice interactions.
- **UI/UX Design:** Material Design 3 guidelines heavily adapted for accessibility (large text, high contrast) suitable for older demographics.

4 Conclusion

SteadySense successfully achieves its goal of providing a reliable, automated safety net for the elderly. By combining hardware sensor analysis with cloud synchronization and AI, the application demonstrates the potential of modern mobile technology to solve real-world healthcare challenges. Future iterations of the app aim to include smartwatch integration and more granular health data analytics.